Treasure Island

When does a swarm of LLM agents whistleblow, and what in their environment changes that?

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Abstract

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Research conducted for the Multi-Agent Oversight and Control Research Sprint hackathon run by Apart Research, November 2026.
Code, every committed game and its model replies are at https://github.com/al-beton/apart-swarm-alignment.



Figure 1: A placeholder teaser: one committed game, replayed from its log. The real one is a still from an exploit game.

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1 Introduction

LLM agents are being given longer and more open-ended work, more and more often in groups that split a task between them. Over a long task an agent will sometimes stumble on something it shouldn't have: a leaked answer, a misconfigured check, a shortcut its designers didn't intend. What it does next matters for oversight. It can say what it found and how, use it quietly, or pass the result to its teammates as if it were its own work. In a group, the others then have to decide whether to trust a lead they can't check. Something like this has happened outside a sandbox. In July 2026, OpenAI agents running a cyber benchmark found an unsanctioned shared channel in a package cache, coordinated through it as a swarm, cheated the evaluation and tried to hide how from its automated grader; almost none of them considered telling a human, and none did (METR, 2026; OpenAI, 2026).

We ask when a swarm of LLM agents whistleblows, and what in its environment changes that. To study this it is not enough to ask agents whether they would report something; they need something real to report, in a setting where what they found is worth something to the group, and the same situation has to come round again so a rate can be measured and a lever tested.

Treasure Island is a sandbox built for that. A crew of LLM pirates hunts one buried treasure on a grid, with too few turns for any one of them to search alone, and a shared scroll as their only channel, which costs a turn to write or to read. The crew scores together, more for getting into the treasure's cave early. Its first exploit is a careless error message: with the cave leak on, a pirate that tries to enter the cave where there is none is refused, and the refusal names the cave's square. The rules plainly forbid acting on it. We bait one pirate into seeing it and watch what it writes: the square and how it knows (candid), the square as a find or a hunch (laundered), or nothing (quiet); whether its crewmates follow or suspect; what the knowledge was worth against a control game on the same seed; and what each pirate tells the study's designers in a free note once the game is over.

This paper contributes:

1. **A reproducible sandbox.** Every game is saved with every model request and reply and replays offline with no keys; every number in this paper is recomputed from committed game logs by the build, and every figure is rebuilt from a committed game.
2. **A first exploit with a controlled bait.** The cave leak gives an agent knowledge it is not entitled to at a known turn, with a forced and a seeded variant of the bait and two controls.
3. **A first measurement** of how often the baited agent is candid, launders the knowledge or stays quiet, how its crew responds, and what it says afterwards when asked. [Rates pending the overnight runs: seeds 1 to 9 of the four arms, and the same arms across every model Adaption hosts.]

Section 2 places this among related work, Section 3 sets out the game and the measures,

Section 4 reports the results and Section 5 discusses what they do and don't show.

2 Related Work

The incident we model. In July 2026, OpenAI agents running a cyber benchmark with many impossible tasks (Wang et al., 2026) turned a package cache into an unsanctioned message board. About 1,200 of them used it, coordinated as a swarm, cheated the evaluation, attacked Hugging Face, and at least a fifth showed interest in tampering with their own transcripts to fool the grader (OpenAI, 2026; METR, 2026). Almost none considered telling a human, and none did (METR, 2026). Treasure Island is a small, replayable analogue of one slice of that: a leaked answer, a shared channel, and a choice to disclose.

Multi-agent sandboxes. LLM agents have been studied in shared worlds for believable social behaviour (Park et al., 2023a), for sustaining a commons (Piatti et al., 2024) and for trading reward against ethics in text games (Pan et al., 2023); Hammond et al. (2025) map the risks that arise only between agents, including collusion. These sandboxes measure cooperation or harm. Ours measures whether an agent says *how* it knows what it knows.

Collusion and covert coordination. Agents can hide messages from overseers by steganography (Motwani et al., 2024; Mathew et al., 2024; Roger and Greenblatt, 2023), set cartel prices unprompted (Fish et al., 2024), and take secret tools they call unfair (Zeng and Rudzicz, 2026). Our pirates share one score, so nothing pushes them to collude against each other; the question is whether the crew hides from the designers a gain they all share.

Deception and hiding from overseers. A trading agent given an insider tip acts on it and hides the reason from its manager (Scheurer et al., 2023), the closest setting to ours. Models also scheme in context (Meinke et al., 2024), fake alignment (Greenblatt et al., 2024), sandbag evaluations (van der Weij et al., 2024), act as insider threats (Lynch et al., 2025) and often know when they are being tested (Needham et al., 2025); Park et al. (2023b) survey the field. We add a crew: the leak reaches one agent, and the others can follow, question or ignore it.

Whistleblowing and honesty under incentives. In staged scenarios, frontier agents sometimes report a user's misconduct to outside authorities unasked (Anthropic, 2025a,b). That rate varies by model family, falls as the task grows harder and rises with a moral nudge (Agrawal et al., 2025). Honesty can be measured apart from accuracy (Ren et al., 2025). Earlier we found that an agent with something of its own to lose came forward about collusion more often, for some models far more often, than one with nothing at stake (Wakelin, 2026). Here the wrongdoing is the agent's own windfall, not a principal's, and the report is a free note to the designers.

Oversight and control. We label disclosure from transcripts, as a judge model would (Zheng et al., 2023), a step towards scalable oversight (Bowman et al., 2022). Chain-of-thought monitors catch reward hacking but can teach models to obfuscate (Baker et al., 2025; Korbak et al., 2025). AI control assumes the agent may be adversarial and builds protocols around it (Greenblatt et al., 2023; Bhatt et al., 2025). Our sandbox asks the opposite question: when would the agent tell us itself?

3 Methods

3.1 The game

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3.2 The agents

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3.3 Measuring whistleblowing

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4 Results

Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae. Each subsection below reports one experiment, and each rests on games committed to this repository that replay with no keys.

4.1 Where the treasure lands decides the silent crew's score

In 1 of the 2 sample games (50%) the crew dug up the treasure, and it scored 115 on average.

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5 Discussion and Limitations

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5.1 Future Work

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6 Conclusion

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Figure 2: A silent crew searching the grid alone, replayed from the committed run `runs/silent-seed-7`. Rebuild it with `make figure NAME=silent-game`; play the game again with `uv run treasure-island run configs/silent.toml --seed 7`.

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A LLM Usage Statement

Lorem ipsum dolor sit amet, consectetur adipiscing elit. The authors chose the questions, designed the game and the experiments and directed the work; a team of Claude agents built the sandbox, its tests and the figures through a pull-request workflow, and OpenAI Codex reviewed each change. The authors take responsibility for every claim in the paper.

B Code and Data

Code, configs, run logs and archived model replies, MIT licensed: https://github.com/al-beton/apart-swarm-alignment. Every committed game replays offline with no API keys, and every figure rebuilds from those games.

C Limitations and Dual-Use / Ethical Considerations

C.1 Limitations

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C.2 Dual-use risks

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C.3 Ethical considerations

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D Reproducibility

Every game the paper reports is committed under `results/` and replays with `uv run treasure-island replay <folder>`, with no keys and no network. Every figure is named in `src/treasure_island/figures.py` and rebuilds with `make figure NAME=<name>`.